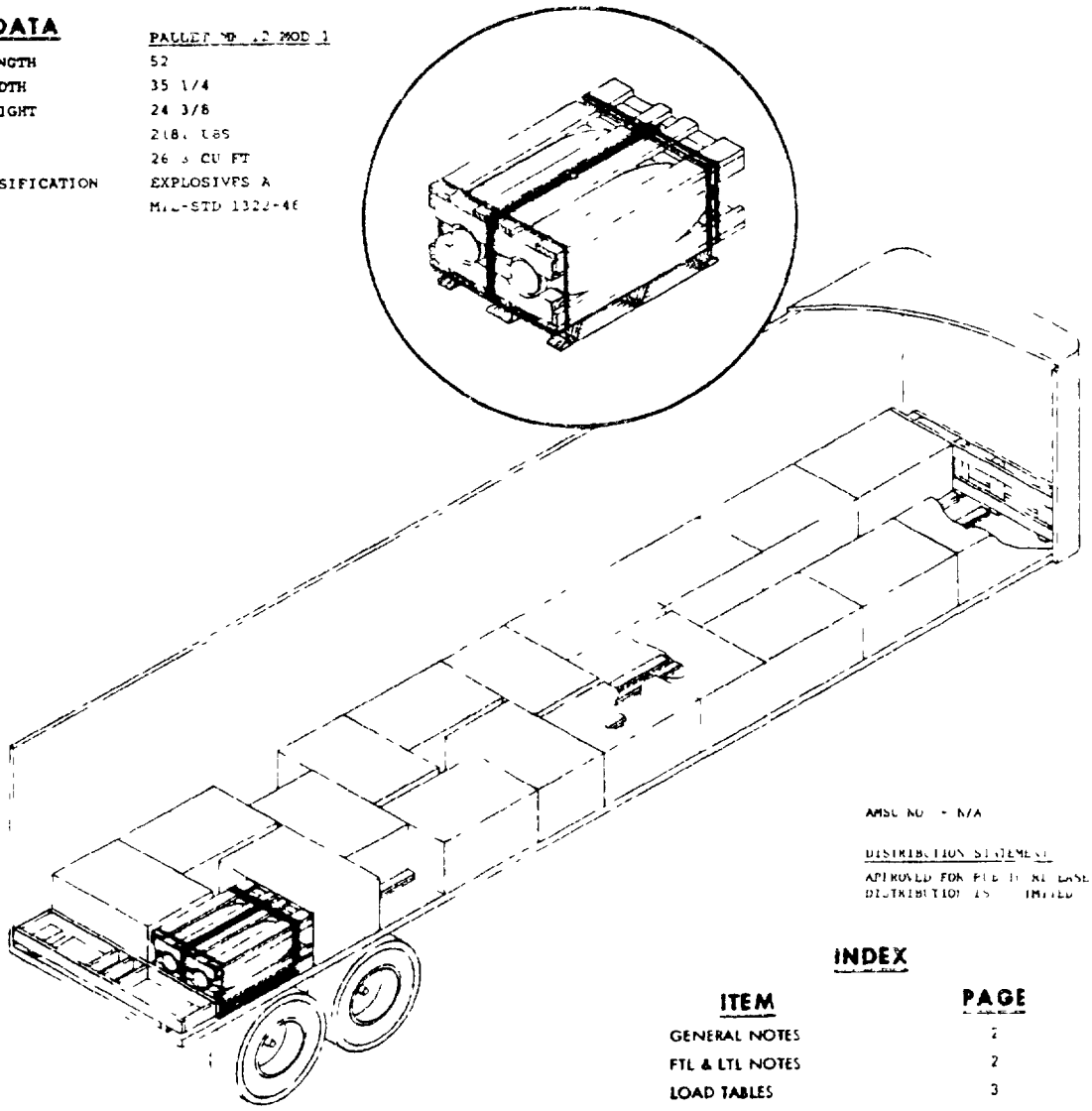
MILITARY STANDARD

MIL-STD-1320-174A
(NAVY)
11 JANUARY 1990
SUPERSEDING
MIL-STD-1320-174
20 APRIL 1979

TRUCKLOADING
WARHEADS, GUIDED MISSILE, MK 40 MOD 0
AND WDU-25/B
DOMESTIC UNIT LOAD

UNIT LOAD DATA

	PALLET MX-2 MOD 1
DIMENSIONS - LENGTH	52
WIDTH	35 1/4
HEIGHT	24 3/8
WEIGHT (APPROX)	2181 LBS
CUBE	26.3 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES A
UNIT LOAD DWG	MIL-STD 1320-46



AMSC NO. - N/A

DISTRIBUTION STATEMENT

APPROVED FOR PUBLIC RELEASE
DISTRIBUTION IS UNLIMITED

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REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	
A	ADDED TO TITLE "AND WDU-25/B	1-11-90	EK	KT

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE 19DEC

SIGNATURE SEA SYSCOM BY DIRECTION DATE

ORIGINATOR

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1 THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING DOMESTIC AND OVERSEAS UNIT LOADS OF THE MK 40 GUIDED MISSILE WARHEAD THE UNIT LOAD SHALL BE IN ACCORDANCE WITH MIL-STD-1322-46

2 THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42 44 AND 45 FT TRAILERS WITH NAILABLE FLOORS THE TRAILER AXLES SHALL BE LOCATED IN THE "WESTERN" POSITION (AT EXTREME REAR OF THE TRAILER)

3 WHEN THE SPACE AT THE REAR OF THE LOAD IS LESS THAN 12 INCHES, USE DETAIL C, PAGE 4 FOR THE REAR BLOCKING WHEN THE SPACE IS BETWEEN 12 AND 36 INCHES USE DETAIL B, PAGE 4 DO NOT USE TRAILERS WITH ROLL UP DOORS WHEN BLOCKING BY THIS METHOD WHEN THE SPACE IS GREATER THAN 36 INCHES USE DETAIL G, PAGE 6 FOR REAR BLOCKING

4 THE REAR BLOCKING ASSEMBLIES, DETAIL B OR C, SHALL FILL THE SPACE BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED

5 THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS

6 AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO THE INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS ATTACH EXPLOSIVES A PLACARD TO BOTH SIDES , FRONT AND BACK OF THE TRAILER

7 APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER, MM-L-751 NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT STRAPPING QQ-S-781, TYPE I, CLASS A PLYWOOD, COMMERCIAL STANDARD CS 45, GRADE INTERIOR C-D

8 FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY)

FTL AND LTL 40, 42, 44 & 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES BEFORE LOADING MORE THAN 19 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES FOR TABLES I, II & III ARE MET

1 TABLES I, II AND III PROVIDE THE CORRECT LOAD PATTERNS FOR THE LENGTH OF THE TRAILER BEING LOADED AND THE ALLOWABLE GROSS TANDEM AXLE WEIGHT AND GROSS VEHICLE WEIGHTS THE MAXIMUM NUMBER OF UNIT LOADS THAT CAN BE LOADED INTO A TRAILER IS 40 FT TRAILER - 20 UNIT LOADS
42 FT TRAILER - 21 UNIT LOADS
44/45 FT TRAILER - 22 UNIT LOADS

2 TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR THE LENGTH OF THE TRAILER BEING LOADED OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED, FIND THE CORRECT LOAD PATTERN MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE TRACTOR-TRAILER WEIGHT AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5 & 6

NOTE

LIGHTWEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM
STANDARD WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 7 800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11 500 LBS MAXIMUM

THE ABOVE MAXIMUM WEIGHTS MAY BE ADJUSTED WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET

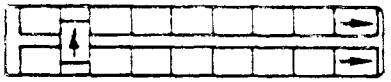
3 THE TRUCKLOAD SHOWN ON PAGE 5 DOES NOT LIMIT SHIPMENT TO THE NUMBER OF UNIT LOADS SHOWN IT IS PRESENTED TO SHOW THE BASIC PRINCIPLES OF BLOCKING ONLY THE LOAD PATTERN TABLES SHOW THE CORRECT NUMBER OF UNIT LOADS FOR THE LENGTH AND WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT RESTRICTIONS OF THE ROUTING

4 WHEN SHIPPING LESS THAN THE AMOUNTS SHOWN IN THE LOAD TABLES, FOLLOW THE PROCEDURES SHOWN ON PAGE 6 DO NOT USE TRAILERS WITH NON-NAILABLE FLOORS

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TABLE I

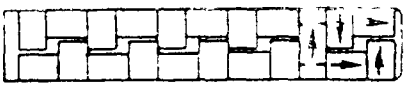
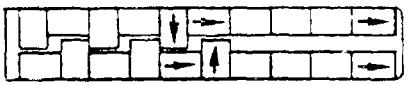
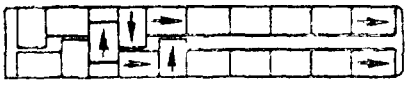
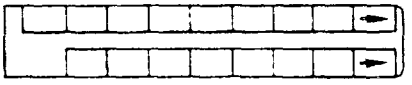
40 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD →)
* 20	
19	
18	
17	
16	
15	

14 OR LESS CONTINUE DOUBLE ROW PATTERN

TABLE II

42 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD →)
* * 21	
* 20	
19	
18	
17	

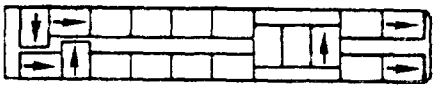
16 OR LESS CONTINUE DOUBLE ROW PATTERN

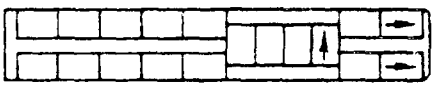
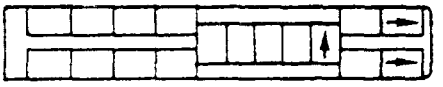
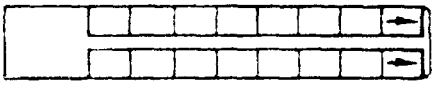
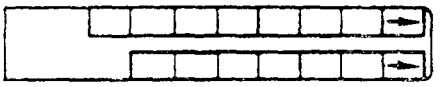
NOTE

ARROW INDICATES NOSE END OF WARHEAD

TABLE III

44 & 45 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN (FWD →)
* * * 22	
* * 21	
* 20	
19	

NO. OF UNIT LOADS	LOAD PATTERN (FWD →)
18	
17	
16	
15	

14 OR LESS CONTINUE DOUBLE ROW PATTERN

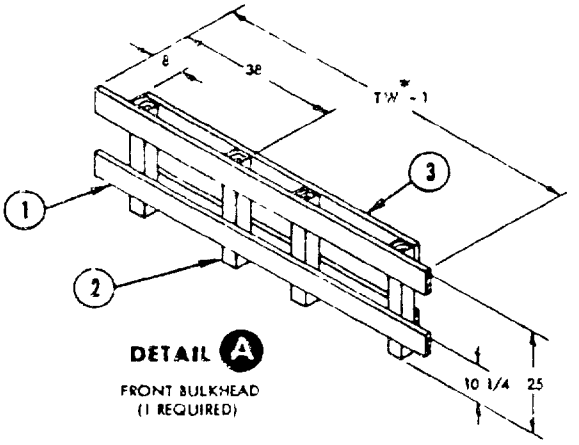
* * * AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED

* * AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED

* AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHTS OF 73,280 LBS AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS

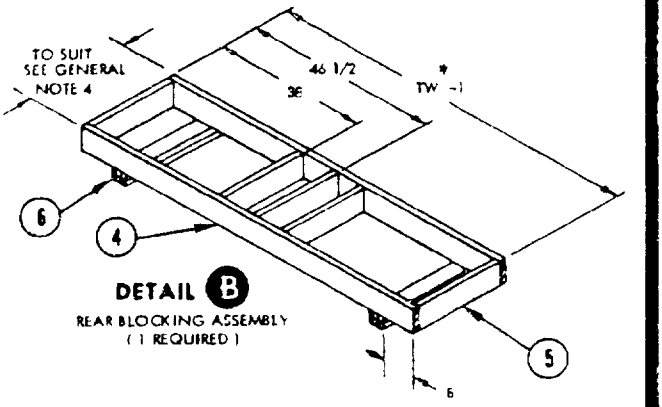
NOTE BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 50 STATES WITHOUT EXCEEDING THE WEIGHT LAWS WHEN THE PATTERN NUMBER IS NOT MARKED WITH AN ASTERISK (*) OR ASTERISKS

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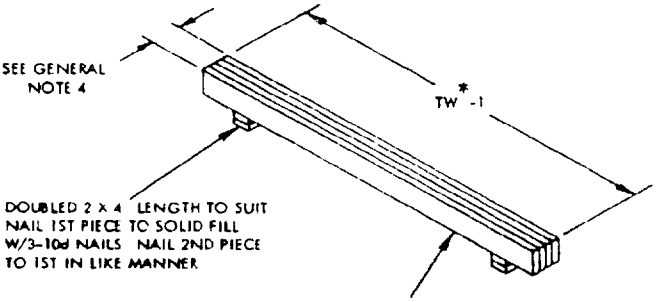
DETAIL A

FRONT BULKHEAD
(1 REQUIRED)



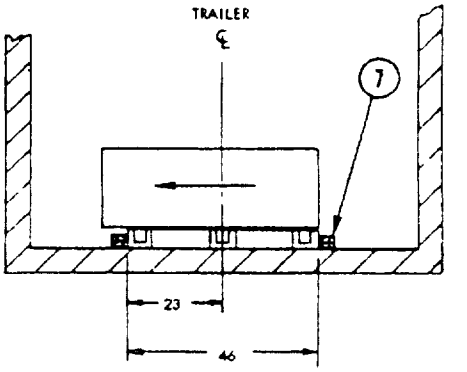
DETAIL B

REAR BLOCKING ASSEMBLY
(1 REQUIRED)



DETAIL C

REAR BLOCKING ASSEMBLY
(1 REQUIRED)
(USED WHEN SPACE AT REAR
IS LESS THAN 12 INCHES)



SECTION D-D

* TW = TRAILER WIDTH

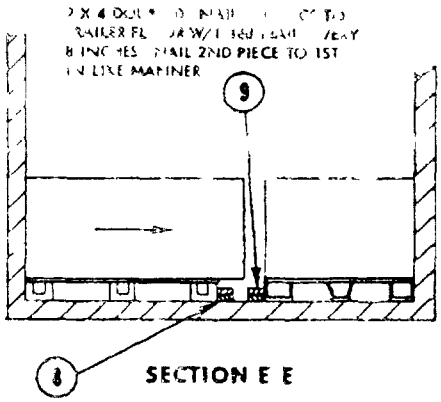
TABLE IV
TRUCKLOAD DATA

NO OF UNIT LOADS	WEIGHT (LBS) APPROX	
	UNIT LOAD W/MK 12 - 0 PALLET	UNIT LOAD W/MK 12 - 1 PALLET
22	48 880	48,070
21	46,620	45,885
20	44, 400	43,700
19	42,180	41,515
18	39,960	39,330
17	37,740	37,145
16	35,520	34,960

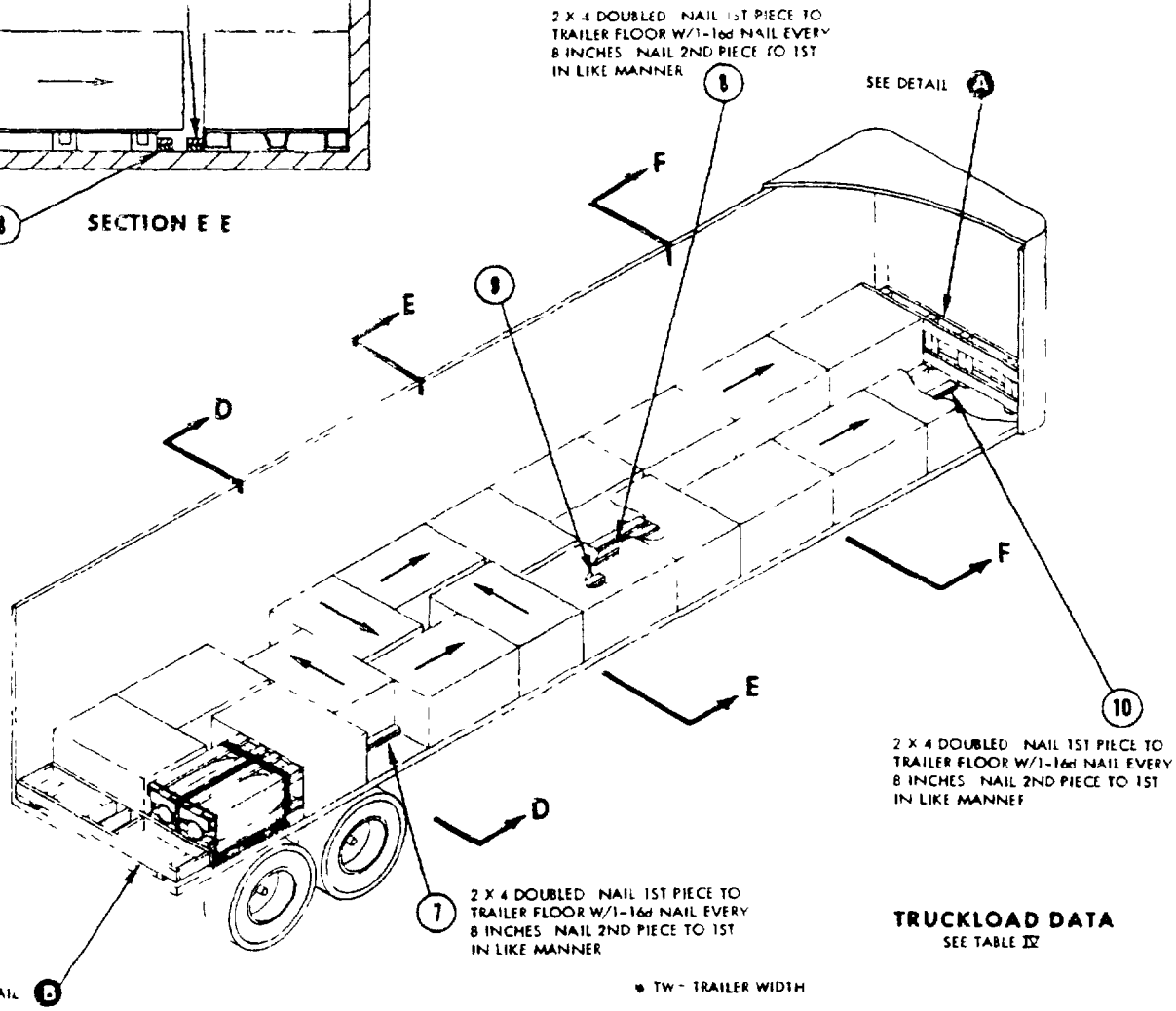
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NOTE

ARROW INDICATES NOSE END OF WARHEAD

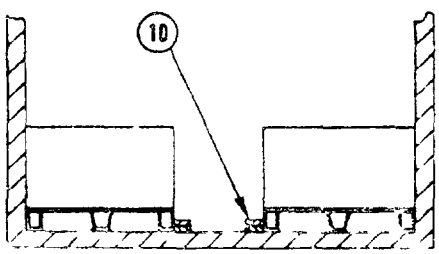


SECTION E E



TRUCKLOAD DATA
SEE TABLE IX

* TW - TRAILER WIDTH



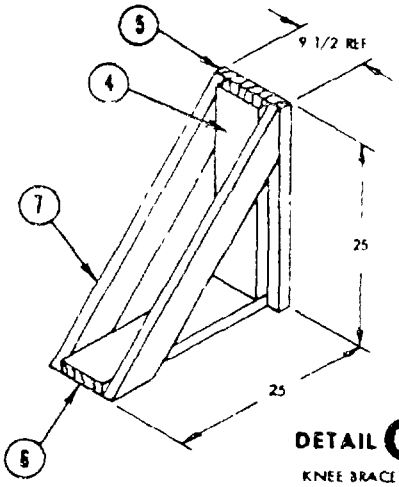
SECTION F-F

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	SLEEPER (DBL ROW)	2 X 4 X 52	12	SEE FIELD NOTE		
9	SLEEPER (CHIMNEY)	2 X 4 X 45	12	SEE SECTION E - E		
8	SLEEPER (CHIMNEY)	2 X 4 X 35	12	SEE FIELD NOTE		
7	SLEEPER (SINGLE ROW)	2 X 4 X 35	4	SEE FIELD NOTE		
6	SUPPORT	2 X 4 TO SUIT	4	4 6	2 PER JOINT 4	16d 10d
5	STRUT	2 X 6 TO SUIT	5	SEE 4	-	-
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3 PER JOINT	16d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3 PER JOINT	16d
2	VERTICAL	4 X 4 X 25	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3 PER JOINT	16d

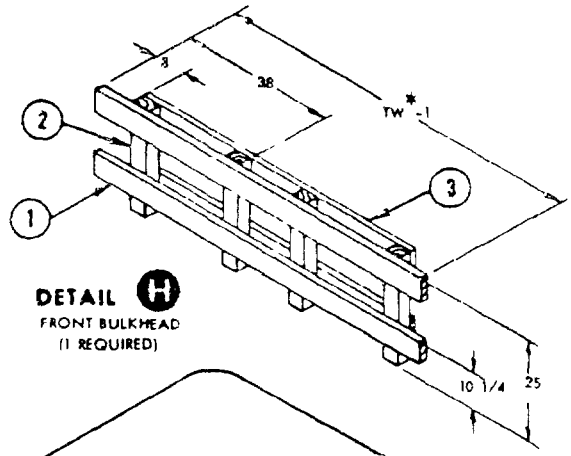
LIST OF MATERIALS & NAILING DATA

TYPICAL LOADING REQUIREMENTS
(19 UNIT LOADS IN 40 FT VAN)

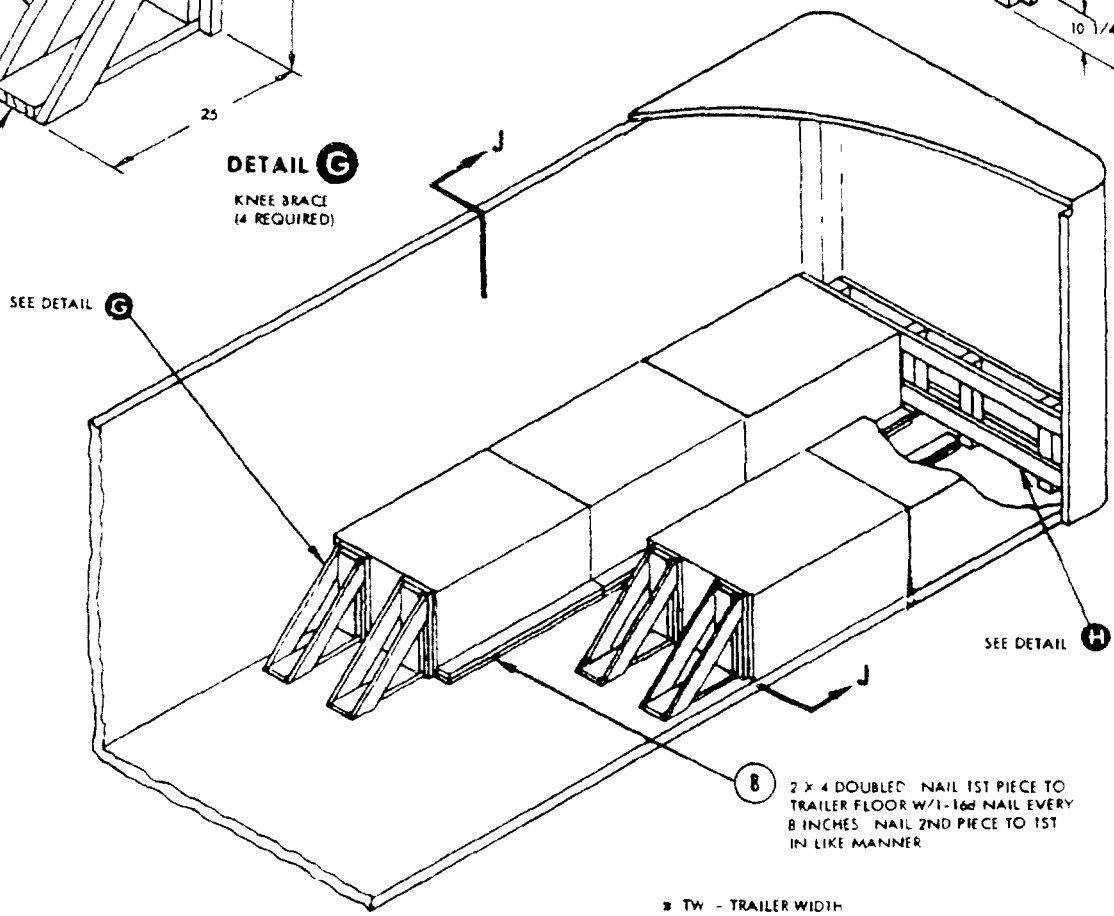
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DETAIL G
KNEE BRACE
(4 REQUIRED)

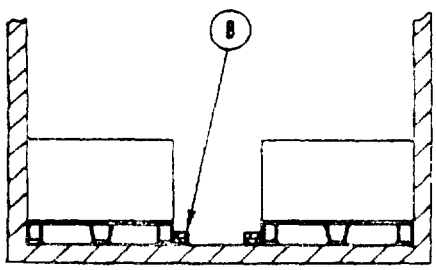


DETAIL H
FRONT BULKHEAD
(1 REQUIRED)



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

* TW - TRAILER WIDTH



SECTION J-J

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
8	SLEEPER	2 X 4 X 52	10	SEE FIELD NOTE		
7	BRACE, DIAGONAL	2 X 6 TO SUIT	8	4 & 6	4/JOINT	16d
6	BRACE, CLEAT	2 X 6 X 23 1/2	4	TRAILER FLOOR	6	16d
5	BRACE, BACKUP	2 X 10 X 25	4	6	3	16d
4	BRACE, UPRIGHT	2 X 6 X 23 1/2	4	5	4	8d
3	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	16d
2	VERTICAL	4 X 4 X 25	4	SEE 1 & 3	-	-
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	16d

LIST OF MATERIALS & NAILING DATA

TYPICAL DOUBLE ROW LTL

REVIEW ACTIVITY
NAVY-OS

PREPARING ACTIVITY
NAVY-OS
(PROJECT NO 8140-N859)

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MIL-STD-1320-174A - Truckloading f/W/H's GM MK 40 Mod 0 & WDU-25/B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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